

Setting Up an offensive Mini-Lab with Ludus

 bu5h.medium.com/setting-up-an-offensive-mini-lab-with-ludus-9859bfca6bc2

bu5h

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Quick Overview

In this guide, I'll walk you through setting up a compact yet powerful mini PC lab using a Beelink SER8 with 32GB RAM and a 1TB SSD. We'll install Debian 12 as our base OS, get Ludus running for automating lab setups, and build a versatile cybersecurity environment with different virtual machines. Whether you're learning or testing, this affordable setup has you covered.

1. What Powers This Lab?

At first, I thought I'd have to build a custom rig from scratch to get the power and flexibility I wanted for my lab. But then I came across the **Beelink SER8**, and it changed my mind. I went with the model that comes with **32GB of RAM and a 1TB SSD** — a bit of a muscle and space to run some virtual machines smoothly. The SER8 is powered by an AMD Ryzen™ 7 8845HS processor, which combines CPU, GPU, and even a Neural Processing Unit (NPU) all in one chip.



Beelink SER8 AMD Ryzen™ 7 8845HS

2. Choosing Ludus

Ludus comes with tons of built-in tools and automation (packer and ansible) that take the hassle out of setting up complex labs, especially when you want to try out new scenarios or attack simulations.

Ludus looks very nice and all, but I won't sugarcoat it: it's not always easy to get jiggy with Ludus at first. The setup can feel a bit tricky and there's a learning curve. But trust me, it's totally worth it once everything starts to click and the pieces begin to align.

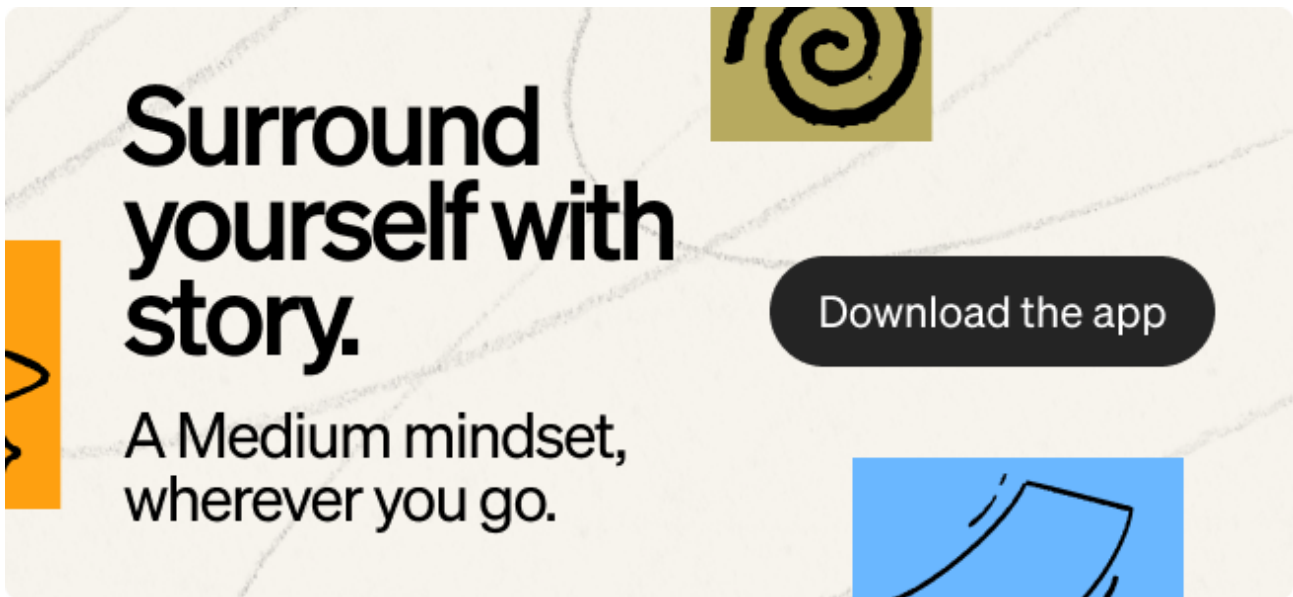
For example, setting up an Active Directory domain can take less than 5 minutes once your environment is ready — a huge time saver compared to doing it all manually.

3. Getting Started with Debian 12

Debian 12 serves as the base OS for this lab. It's stable, well-supported, and resource-efficient.

Installation steps:

1. Download Debian 12 netinst ISO from the official site.
2. Create a bootable USB with Rufus or balenaEtcher.
3. Boot the device from USB and follow the guided install:
 - Use guided partitioning with the entire disk.
 - Select a minimal install (no desktop environment needed).
 - Configure networking with static IP (.....).
4. Complete the install and reboot into Debian.



Once done , Update the box

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```
bu5h@ludus:~$ su root
Password:
root@ludus:/home/bu5h# cd ~
root@ludus:~# apt update && apt install curl sudo
Hit:1 http://security.debian.org/debian-security bookworm-security InRelease
Hit:2 http://deb.debian.org/debian bookworm InRelease
Hit:3 http://deb.debian.org/debian bookworm-updates InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
Reading package lists... Done
Building dependency tree... Done
```

Install ludus

```
curl -s https://ludus.cloud/install | bash
```

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```
root@ludus:~# curl -s https://ludus.cloud/install | bash
=====
LUDUS
=====
[+] Client install prefix set to /usr/local/bin
[+] Created temp dir at /tmp/ludus-client.vXJ6H8
[+] Architecture detected as x86_64
[+] OS detected as Linux
[+] Downloaded ludus-client_linux-amd64-1.9.2 into /tmp/ludus-client.vXJ6H8
[+] Downloaded ludus checksums file into /tmp/ludus-client.vXJ6H8
[+] Checksum of /tmp/ludus-client.vXJ6H8/ludus-client_linux-amd64-1.9.2 verified
[+] Install prefix already exists. No need to create it.
[+] Installed ludus-client_linux-amd64-1.9.2 to /usr/local/bin/ as 'ludus'
[+] Ludus client installation complete
[+] Shell completions already installed
[?] Would you like to install the Ludus server on this host?
[?] (y/n): y
[+] Installing Ludus server

[+] Downloaded ludus-server-1.9.2 into /tmp/ludus-client.vXJ6H8
[+] Checksum of /tmp/ludus-client.vXJ6H8/ludus-server-1.9.2 verified
Ludus server 1.9.2+0d346b7
Extracting ludus to /opt/ludus...
Ludus files extracted successfully

Only run Ludus install on a clean Debian 12
machine that will be dedicated to Ludus

I Understand  Bail
```

Inline installation of ludus

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```
Ludus Interactive Installer //////////////////////////////////////

What is the gateway IP for this host?
The suggestion is usually correct
> 192.168.2.1

What is the netmask for interface enp1s0
The suggestion is usually correct
> 255.255.255.0

What pool will store VMs
The default is 'local'
> Datacenter

In what format should VMs be stored?
Use 'qcow2' for dir storage (local), 'raw' for ZFS
> qcow2

Server Settings
proxmox_node: ludus
proxmox_interface: enp1s0
proxmox_local_ip: 192.168.2.2
proxmox_public_ip: 192.168.2.2
proxmox_gateway: 192.168.2.1
proxmox_netmask: 255.255.255.0

shift+tab back • enter next //////////////////////////////////////
```


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```
Ludus Interactive Installer //////////////////////////////////////

What pool will store ISO files?
The default is 'local'
> local

What new interface will be used to NAT out of enp1s0
Default is 'vmbr1000' - don't change without a reason
> vmbr1000

Should users be allowed to add ansible roles?
Admins can always add ansible roles

Deny Allow

Server Settings
proxmox_node: ludus
proxmox_interface: enp1s0
proxmox_local_ip: 192.168.2.2
proxmox_public_ip: 192.168.2.2
proxmox_gateway: 192.168.2.1
proxmox_netmask: 255.255.255.0
proxmox_vm_storage_pool:
vDatacenter
proxmox_vm_storage_format: qcow2
proxmox_iso_storage_pool: local
ludus_nat_interface: vmbr1000

←/→ toggle • shift+tab back • enter next • y Deny • n Allow //////////////////////////////////////
```

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```
Ludus Interactive Installer //////////////////////////////////////

Expose the admin API globally?
Default is 'deny' to listen on 127.0.0.1:8081

Allow Deny

Do you have a Ludus license key?
Leave 'community' for community edition
> community

Server Settings
proxmox_node: ludus
proxmox_interface: enp1s0
proxmox_local_ip: 192.168.2.2
proxmox_public_ip: 192.168.2.2
proxmox_gateway: 192.168.2.1
proxmox_netmask: 255.255.255.0
proxmox_vm_storage_pool:
vDatacenter
proxmox_vm_storage_format: qcow2
proxmox_iso_storage_pool: local
ludus_nat_interface: vmbr1000
prevent_user_ansible_add: false

shift+tab back • enter next //////////////////////////////////////
```

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```
Ready?
Install Wait, no

Server Settings
proxmox_node: ludus
proxmox_interface: enp1s0
proxmox_local_ip: 192.168.2.2
proxmox_public_ip: 192.168.2.2
proxmox_gateway: 192.168.2.1
proxmox_netmask: 255.255.255.0
proxmox_vm_storage_pool:
vDatacenter
proxmox_vm_storage_format: qcow2
proxmox_iso_storage_pool: local
ludus_nat_interface: vmbr1000
prevent_user_ansible_add: false
expose_admin_port: true
license_key: community

←/→ toggle • shift+tab back • enter submit • y Install • n Wait, no //////////////////////////////////
```

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```
• Ludus install will cause the machine to reboot twice.
• Install will continue automatically after each reboot.
• Check the progress of the install by running:
  'ludus-install-status' from a root shell.

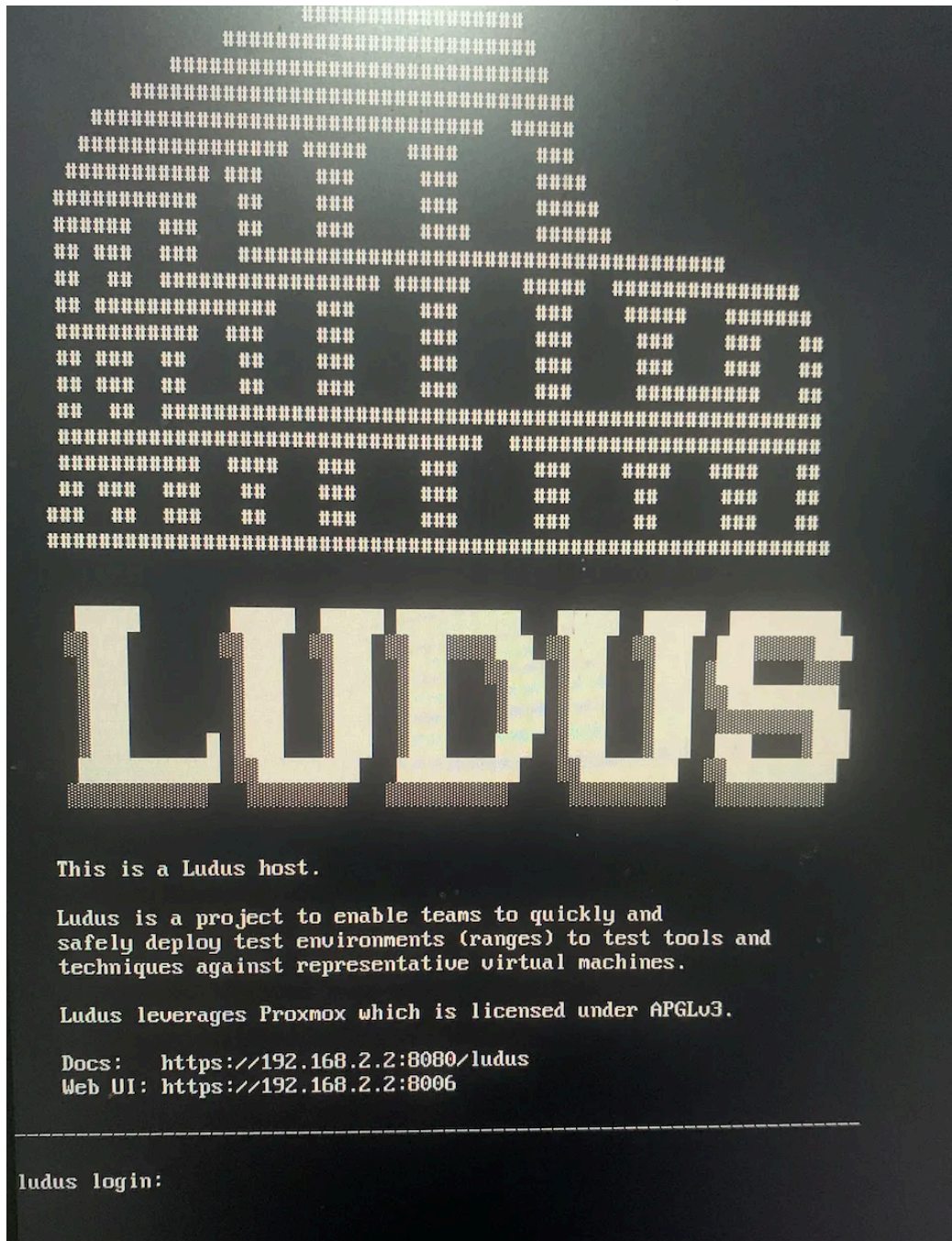
Starting install now!

Installing ansible with pip
Updating apt cache...
Installing python3-pip...

Installing ansible with pip...
```

Installation

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with no ssh , here is to a reboot that successfully went through

```
ludus-install-status
```

Press enter or click to view image in full size



Create a user

```
LUDUS_API_KEY='ROOT.xxxx' \  
ludus user add --name "Admin User" --userid admin --admin --url
```


https://127.0.0.1:8081

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```
INFO] Adding user to Ludus, this can take up to a minute. Please wait.
```

USERID	PROXMOX USERNAME	ADMIN	API KEY
admin	admin-user	true	admin. [REDACTED]

Export the key

```
export LUDUS_API_KEY='admin.xxxx'
```

Then, start interacting with Ludus

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```
root@ludus:~# ludus user creds get
```

PROXMOX USERNAME	PROXMOX PASSWORD
admin-user	[REDACTED]

Pulling the templates , the longest part of the installation

```
~$ git clone https://gitlab.com/badsectorlabs/ludus
```

```
~$ cd ludus/templates
```

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```
root@ludus:~/ludus/templates# ls -l
```

Permissions	Size	Owner	Group	Size	Month	Day	Time	File
-rw-r--r--	1	root	root	527	Apr	7	23:02	README.md
drwxr-xr-x	5	root	root	115	Apr	7	23:02	commando-vm
drwxr-xr-x	4	root	root	57	Apr	7	23:02	debian10
drwxr-xr-x	5	root	root	112	Apr	7	23:02	flare-vm
drwxr-xr-x	2	root	root	35	Apr	7	23:02	manual-setup-required
drwxr-xr-x	4	root	root	72	Apr	13	14:11	remnux
drwxr-xr-x	4	root	root	67	Apr	7	23:02	rocky-8-x64-server
drwxr-xr-x	4	root	root	67	Apr	7	23:02	rocky-9-x64-server
drwxr-xr-x	4	root	root	72	Apr	7	23:02	ubuntu-20.04-x64-server
drwxr-xr-x	4	root	root	72	Apr	7	23:02	ubuntu-22.04-x64-server
drwxr-xr-x	4	root	root	73	Apr	7	23:02	ubuntu-24.04-x64-desktop
drwxr-xr-x	4	root	root	72	Apr	7	23:02	ubuntu-24.04-x64-server
drwxr-xr-x	4	root	root	97	Apr	7	23:02	win10-22h2-x64-enterprise
drwxr-xr-x	5	root	root	112	Apr	7	23:02	win11-23h2-x64-enterprise
drwxr-xr-x	4	root	root	92	Apr	7	23:02	win2012r2-server-x64
drwxr-xr-x	4	root	root	90	Apr	7	23:02	win2016-server-x64
drwxr-xr-x	5	root	root	105	Apr	7	23:02	win2019-server-x64
drwxr-xr-x	4	root	root	110	Apr	7	23:02	win2019-server-x64-no-security-updates

Add a template

```
ludus templates add -d ubuntu-24.04-x64-server
ludus templates list
```

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```
root@ludus:~/ludus/templates# ludus templates add -d ubuntu-24.04-x64-server
[INFO] Successfully added template
root@ludus:~/ludus/templates# ludus templates list
```

TEMPLATE	BUILT
debian-11-x64-server-template	TRUE
debian-12-x64-server-template	TRUE
kali-x64-desktop-template	TRUE
win11-22h2-x64-enterprise-template	TRUE
win2022-server-x64-template	TRUE
flare-vm-template	TRUE
ubuntu-24.04-x64-desktop-template	TRUE
ubuntu-24.04-x64-server-template	FALSE
win2016-server-x64-template	TRUE
win2019-server-x64-template	TRUE

```
root@ludus:~/ludus/templates# exit
```

Build the template

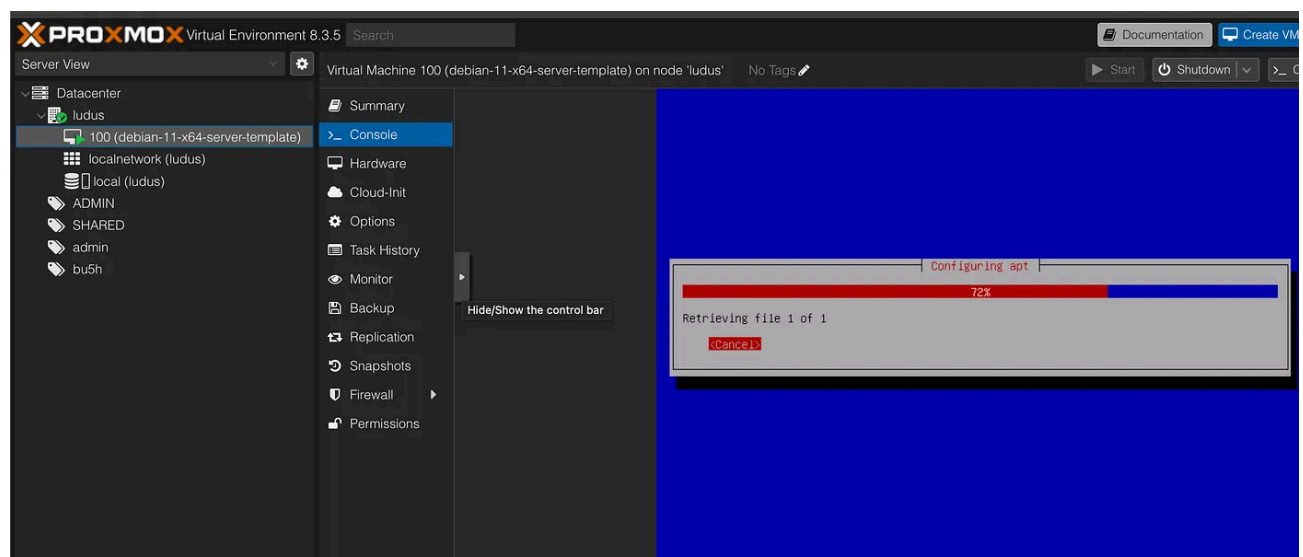
```
ludus templates build -n ubuntu-24.04-x64-server-template
```

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```
bu5h@ludus:~$ ludus templates build -n ubuntu-24.04-x64-server-template
[INFO] Template building started - this will take a while. Building 1 template(s) at a time.
bu5h@ludus:~$ ludus templates logs -f
2025/05/17 12:19:50 ui: ==> proxmox-iso.ubuntu2404-server: Retrieving ISO
2025/05/17 12:19:50 ui: ==> proxmox-iso.ubuntu2404-server: Trying https://releases.ubuntu.com/24.04.2/ubuntu-24.04.2-live-server-amd64.iso
2025/05/17 12:19:50 ui: ==> proxmox-iso.ubuntu2404-server: Trying https://releases.ubuntu.com/24.04.2/ubuntu-24.04.2-live-server-amd64.iso?checksum=sha256%3Ad6dab0c3a657988501b4bd76f1297c053df710e060c3a6c60d6ad24f270b4d
```

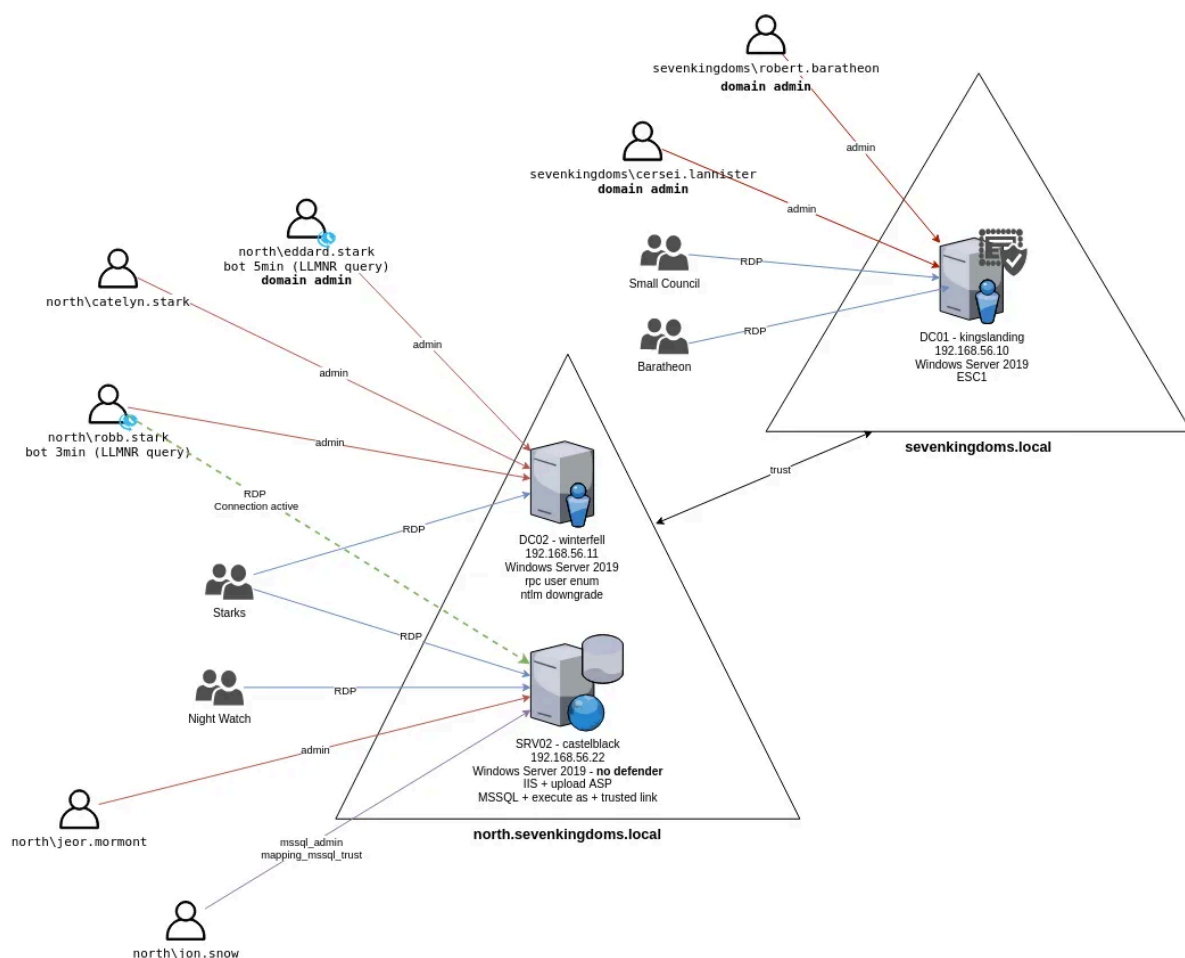
And here we have it

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Setup GOAD-Light from GOAD Project

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```
git clone https://github.com/Orange-Cyberdefense/GOAD.git
```

```
cd GOAD
```

```
sudo apt install python3.11-venv
```

```
./goad.sh
```

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```
GOAD/ludus/local > set_lab GOAD-Light
GOAD-Light/ludus/local > install
[+] Current Settings
[+] Current Lab      : GOAD-Light
[+] Current Provider : ludus
[+] Current Provisioner : local
[+] Extension(s)    :
Create lab with theses settings ? (y/N)y
[+] Create instance folder
[+] Create instance providing files
[+] Instance vagrantfile created : /workspace/eaibd5-goad-light-ludus/provider/config.yml
[+] Create lab provisioning file inventory_disable_vagrant
[+] Lab inventory file created : /workspace/eaibd5-goad-light-ludus/inventory_disable_vagrant
[+] Create instance provisioning files
[+] Instance inventory file created : /workspace/eaibd5-goad-light-ludus/inventory
[+] Create instance extensions inventory files
[+] Instance eaibd5-goad-light-ludus created
[+] Instance eaibd5-goad-light-ludus loaded
```

Instance ID	Lab	Provider	IP Range	Status	Is Default	Extensions
> eaibd5-goad-light-ludus	GOAD-Light	ludus	192.168.56.0/24	not provided	No	

```
[+] Launch providing
[+] Using api key from env
[+] CWD: /workspace/eaibd5-goad-light-ludus/provider
[+] Running command : ludus version
[+] Api key is set
[+] Using api key from env
[+] CWD: /workspace/eaibd5-goad-light-ludus/provider
[+] Running command : ludus --url https://127.0.0.1:8081 user list --json
[{"name": "Admin User", "userID": "admin", "dateCreated": "2025-04-07T21:13:07.851971752+02:00", "dateLastActive": "2025-05-02T00:43:44.670229657+02:00", "isAdmin": true, "proxmoxUsername": "admin-user", "portforwardingEnabled": false}]
```

Provisioning the lab

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```
GOAD-Light/ludus/local (eafbd5-goad-light-ludus) > provision_lab
[*] Loading inventory
[+] Lab inventory : /home/bu5h/GOAD/ad/GOAD-Light/data/inventory file found
[+] Provider inventory : /home/bu5h/GOAD/workspace/eafbd5-goad-light-ludus/inventory file found
[+] Global inventory : /home/bu5h/GOAD/globalsettings.ini file found
[*] Loading playbook list
[+] build.yml file found
[+] ad-servers.yml file found
[+] ad-parent_domain.yml file found
[+] ad-child_domain.yml file found
[+] wait5m.yml file found
[+] ad-members.yml file found
[+] ad-trusts.yml file found
[+] ad-data.yml file found
[+] ad-gmsa.yml file found
[+] laps.yml file found
[+] ad-relations.yml file found
[+] adcs.yml file found
[+] ad-acl.yml file found
[+] servers.yml file found
[+] security.yml file found
[+] vulnerabilities.yml file found
```

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```
PLAY RECAP *****
dc01      : ok=18  changed=6  unreachable=0  failed=0  skipped=2  rescued=0  ignored=0
dc02      : ok=33  changed=15  unreachable=0  failed=0  skipped=6  rescued=0  ignored=0
srv02     : ok=18  changed=10  unreachable=0  failed=0  skipped=1  rescued=0  ignored=0
[*] Lab successfully provisioned in 02:44:32
```

Press enter or click to view image in full size

```
GOAD-Light/ludus/local (eafbd5-goad-light-ludus) > status
[*] Using api key from env
[*] CWD: /workspace/eafbd5-goad-light-ludus/provider
[*] Running command : ludus --user GOADLighteafbd5 range status
```

USER ID	RANGE NETWORK	LAST DEPLOYMENT	NUMBER OF VMS	DEPLOYMENT STATUS	TESTING ENABLED
GOADLighteafbd5	10.5.0.0/16	2025-05-11 19:21	5	Success	FALSE

PROXMOX ID	VM NAME	POWER	IP
112	GOADLighteafbd5-router-debian11-x64	Off	null
113	GOADLighteafbd5-GOAD-DC01	Off	null
114	GOADLighteafbd5-GOAD-DC02	Off	null
115	GOADLighteafbd5-GOAD-SRV02	Off	null

In case you have a Kali VM deployed outside the Range of GOAD, this is how it can be interconnected via the GOAD router

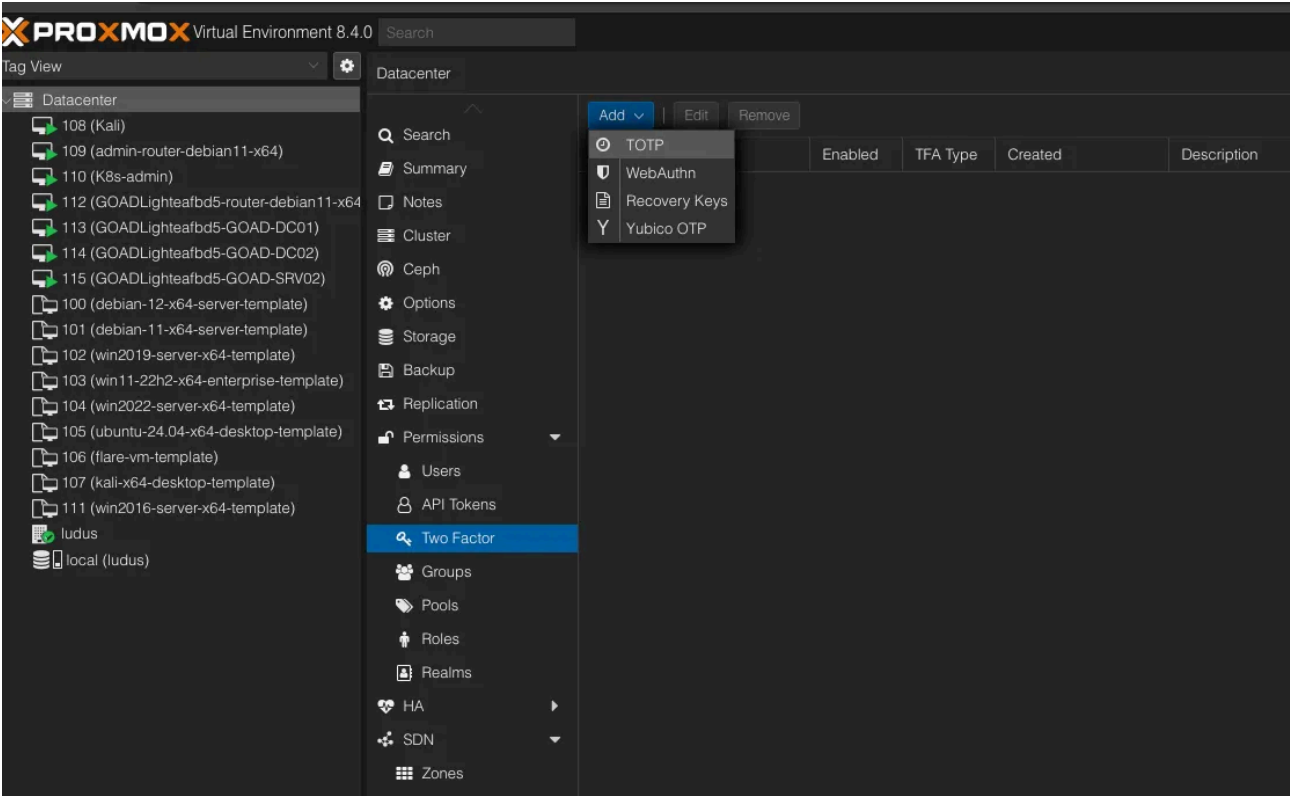
```
iptables -I LUDUS_DEFAULTS -i GOAD_router_interface -s IP_KALI -j ACCEPT
```

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```
root@GOADLighteafbd5-router:~# iptables -I LUDUS_DEFAULTS -i ens18 -s 192.0.2.102 -j ACCEPT
root@GOADLighteafbd5-router:~#
```

Secure your cluster with a TOTP

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Add a TOTP login factor

User:

root@pam

Description:

Google TOTP

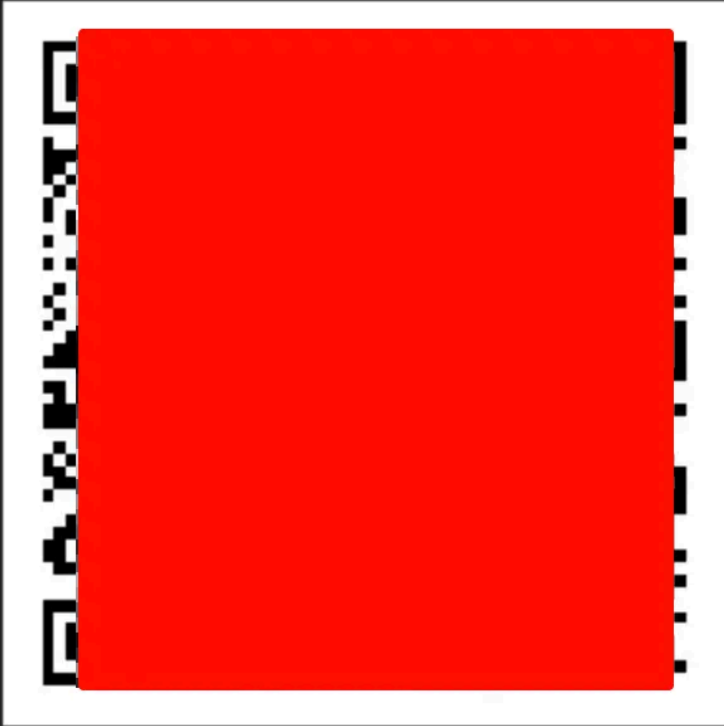
Secret:

V[REDACTED]3

Randomize

Issuer Name:

Proxmox VE - ludus



Verify Code:

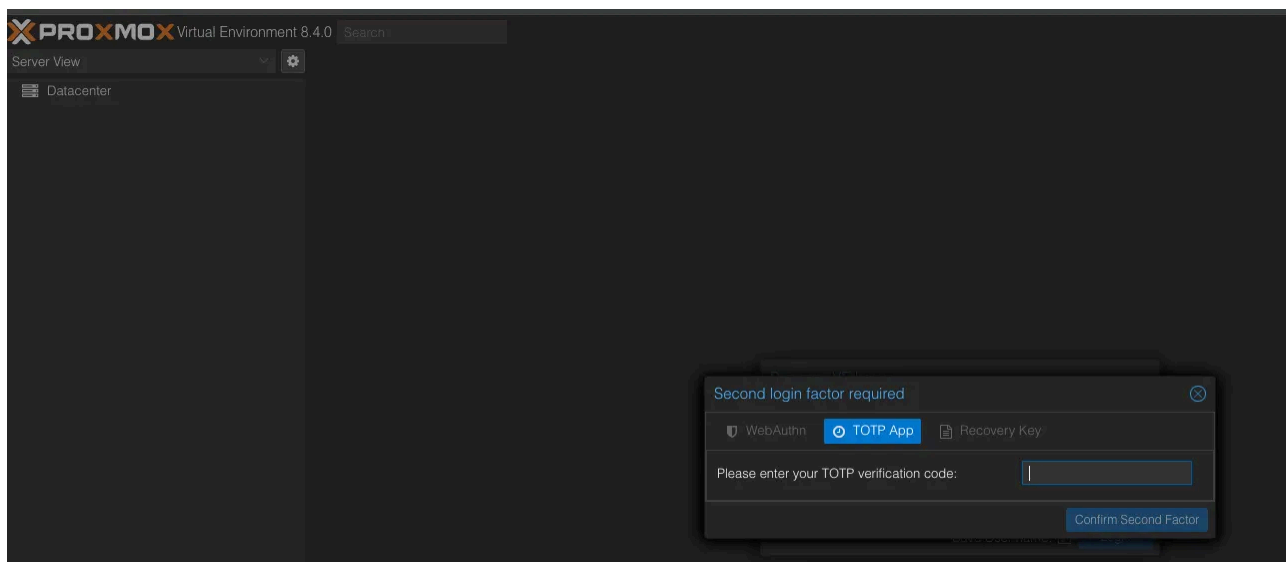
[REDACTED]

Help

Add

Once back at login , you will be prompted for an OTP

Press enter or click to view image in full size



Conclusion

One thing I have to highlight right away: if you're diving into Ludus, **don't skip the Discord community**. It's genuinely one of the best parts of the whole experience. Whether you're stuck during setup, looking to troubleshoot something weird, or just curious about how others are using their labs — the community is active, helpful, and full of people who've already been through the same learning curve.

Ludus takes a bit of getting used to, but once things click, it makes lab building and experimentation genuinely fun. From quick AD deployments to running attack scenarios with different distros, it gives you real control without the usual setup headaches.

My SER8 with 32GB RAM handled things decently, but heavier labs did push it — I occasionally had to pause a VM or two. If you plan to go big, consider upgrading to a **64GB or more**.

Overall, I'm really happy with how everything turned out. It's a quiet, capable little setup that gives me the freedom to tinker, test, and learn.

Thanks for reading! Feel free to reach out if you're building your own lab or just want to chat.

References:

- <https://docs.ludus.cloud/>
- <https://xphantom.nl/posts/Offensive-Security-Lab/#connect-to-your-lab-from-anywhere>
- <https://orange-cyberdefense.github.io/GOAD/providers/ludus/>